Muhammad Shahid

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Education

University of Alberta Edmonton, AB

Bachelor of Science, Computing Science *Expected Apr. 2028*

- **Relevant Coursework:** Software Engineering, Data Structures & Algorithms, Machine Learning, Operating Systems, Database Management, Computer Organization
- **Availability:** Seeking a 4, 8, 12, or 16-month co-op or internship starting May 2027

Technical Skills

Languages: Python, Java, C, JavaScript, SQL, HTML/CSS
Frameworks & Libraries: Flask, React, Android SDK, NumPy, pandas, scikit-learn, Matplotlib, Chart.js, JUnit
Databases & Cloud: PostgreSQL (Supabase), SQLite, Firebase Firestore, Render
Tools & Practices: Git/GitHub, Linux/Bash, Android Studio, REST APIs, OAuth 2.0, Agile/Scrum, UML, Unit Testing

Projects

Ledger – Job Application Tracker | *Python, Flask, PostgreSQL (Supabase), JavaScript* Live Demo

- Built a full-stack job tracker with a drag-and-drop Kanban board (Applied, Interview, Offer, Rejected), persisting status changes through RESTful Flask endpoints
- Migrated the data layer from SQLite to Supabase-hosted **PostgreSQL** with incremental schema migrations, shipping new features without losing existing user data
- Implemented Supabase Auth and **Google OAuth 2.0**, with custom identity resolution allowing sign-in by username or email
- Added deadline tracking with urgency indicators, soft-delete archiving, CSV export, and a mobile-first responsive UI with dark mode

PocketPal – Expense Tracker | *Python, Flask, SQLite, Chart.js* Live Demo

- Developed CRUD expense management with multi-field filtering (date range, category, amount) on a normalized SQLite schema with foreign-key constraints and server-side validation
- Secured accounts with hashed passwords, complexity rules, Google OAuth 2.0 sign-in, and re-authentication before password changes or account deletion
- Built an analytics dashboard with **Chart.js** showing monthly spending trends and category breakdowns

AI Spending Assistant | *React, Flask, LLM API* Live Demo

- Built a **React** chat interface that answers natural-language questions about a user's PocketPal expenses through an LLM API, supporting follow-up questions
- Precomputed all figures (totals, category breakdowns, averages, min/max) in Python and passed them as structured context, so the LLM explains numbers rather than calculating them
- Replaced session cookies with signed, expiring tokens to fix cross-origin login failures caused by third-party cookie blocking; deployed frontend and API as separate services with CORS

Academic Projects

Team Android App | *Java, Android SDK, Firebase Firestore, JUnit, Espresso* *Software Engineering*

- Delivered an Android app in a six-person **Scrum** team over four sprints, using GitHub Projects for sprint planning and pull requests for peer code review
- Turned user stories into UML designs and implemented Firestore-backed features with real-time sync, covered by **JUnit** unit tests and **Espresso** UI tests

Unix Shell & Concurrency | *C, Linux, POSIX Threads* *Operating Systems*

- Implemented a Unix shell in **C** with fork/exec, pipes, I/O redirection, and background job control, plus a multithreaded producer–consumer system debugged with Valgrind and GDB

Regression & Classification from Scratch | *Python, NumPy, scikit-learn, Matplotlib* *Machine Learning*

- Implemented linear and logistic regression with gradient descent in **NumPy**, using L2 regularization and k-fold cross-validation, and benchmarked results against scikit-learn